

Backend Development Field Guide

A competency-oriented guide to reliable backend engineering.

How to use this guide

Use these topics as a study and evidence checklist. Competra does not treat reading a guide as proof of competency; pair learning with practice, a project task and an unfamiliar-context assessment.

1. API design

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

2. Validation

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

3. Database modeling

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

4. Transactions and errors

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

5. Unit and integration testing

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

6. Security basics

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

7. Deployment and operations

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.